SMART INDIA HACKATHON 2026

RaahSetu

AI-based logistics and accessibility intelligence for the Northeast

Problem Statement

SIH26002

Ministry / Organization

Ministry of Development of North Eastern Region

Team Name

RaahSetu

01 IDEA / PROPOSED SOLUTION

Safer logistics decisions when the fastest road is not the best road

The gap

Road disruptions in the Northeast can turn a short route into a delayed or unsafe delivery.

Standard navigation optimizes travel time. Fleet operators also need road condition, accident history, weather exposure and accessibility status.

Observed hazard | Route closure | Weather risk

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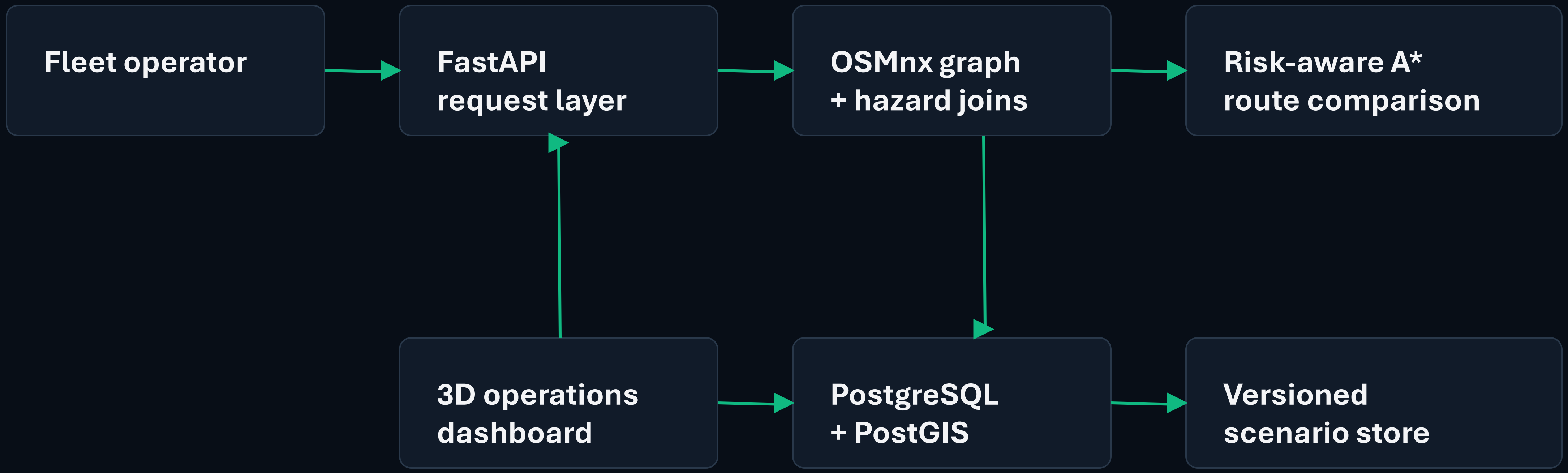
Risk-aware A* routing

A custom cost function adds modelled hazard exposure to travel time, then searches the road graph for a route that balances safety and delay.

Output: route, estimated delay, risk evidence and a clear explanation for operators.

02 TECHNICAL ARCHITECTURE

One explainable path from a request to a route decision



Technology stack

React + TypeScript + React Three Fiber | Python 3.12 + FastAPI | OSMnx | Supabase PostgreSQL + PostGIS | Docker + Kubernetes

03 USE CASES

A common decision layer for emergency and commercial fleets

01

Emergency response

NDRF / SDRF fleets

Plan a relief run around flooded, blocked or landslide-prone segments.

02

Essential logistics

Heavy trucks

Compare time and risk before moving medicines, food or construction material.

03

Field reporting

District officials

Upload a geo-tagged closure or photo so the next route decision uses current evidence.

Operational loop

Request → Compare → Explain → Act

If every road is blocked, the system surfaces the accessibility gap for escalation instead of returning a misleading route.

A realistic pilot plan keeps the decision traceable

Dependencies

01 Open road network from OSM / OSMnx

02 Reviewed hazard observations from government sources

03 Weather and disruption updates

04 Supabase project with PostGIS enabled

Show stoppers and response

Paid live traffic feeds	Use open graph data and expose freshness clearly
Unverified hazard coordinates	Keep records reviewable before promoting them to route cost
Single-access road blocked	Raise an accessibility alert and show the last feasible plan
Low-connectivity field update	Queue geo-tagged reports for offline sync

Submission check 5 slides | no college name or logo | no team photos | export to PDF before portal upload